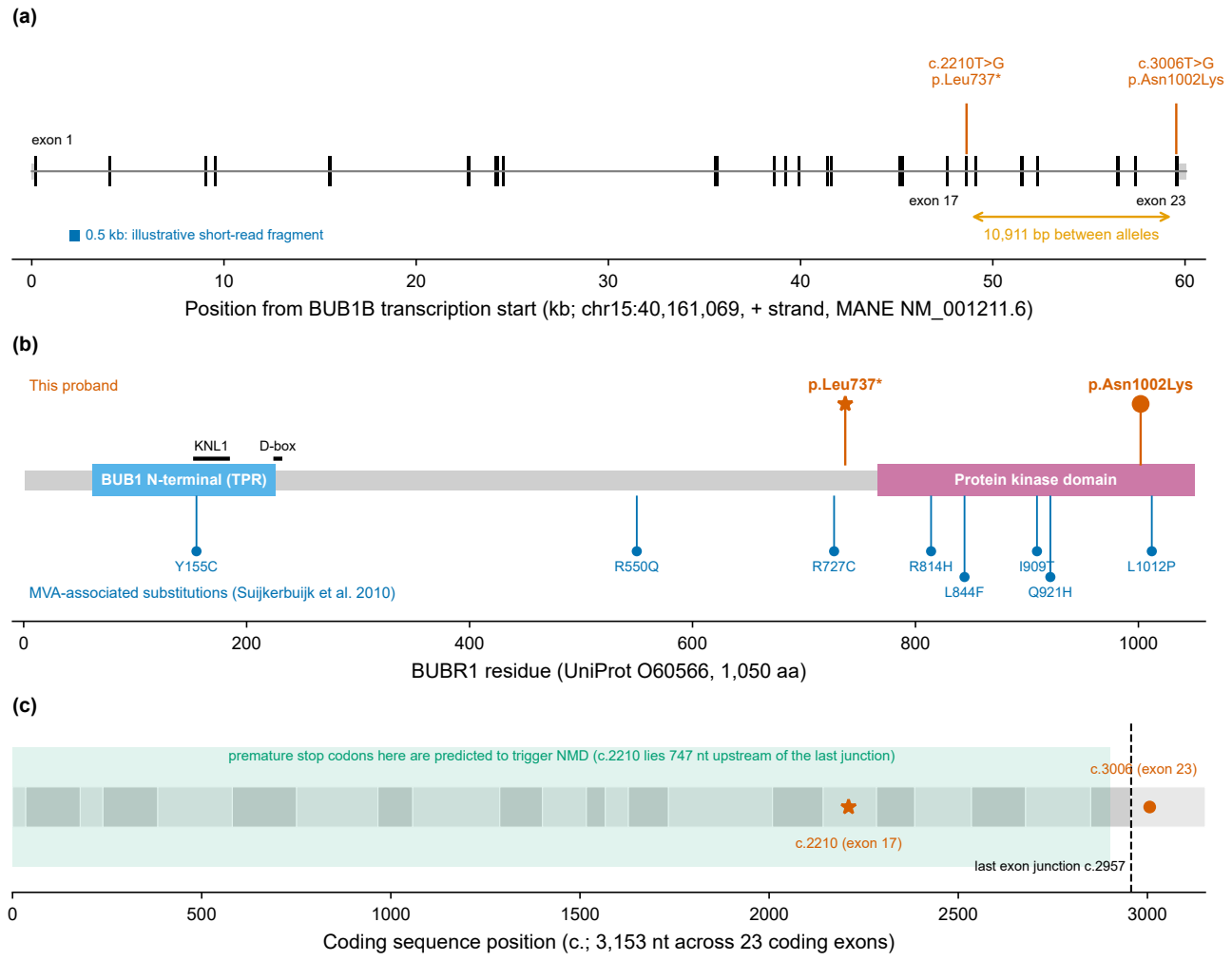

(a)
c.2210T>G
p.Leu737*
c.3006T>G
p.Asn1002Lys
exon 1
exon 17
exon 23
0.5 kb: illustrative short-read fragment
10,911 bp between alleles
0
10
20
30
40
50
60
Position from BUB1B transcription start (kb; chr15:40,161,069, + strand, MANE NM_001211.6)
(b)
This proband
p.Leu737*
p.Asn1002Lys
KNL1
D-box
BUB1 N-terminal (TPR)
Protein kinase domain
Y155C
R550Q
R727C
R814H
L844F
I909T
Q921H
L1012P
MVA-associated substitutions (Suijkerbuijk et al. 2010)
0
200
400
600
800
1000
BUBR1 residue (UniProt O60566, 1,050 aa)
(c)
premature stop codons here are predicted to trigger NMD (c.2210 lies 747 nt upstream of the last junction)
c.3006 (exon 23)
c.2210 (exon 17)
last exon junction c.2957
0
500
1000
1500
2000
2500
3000
Coding sequence position (c.; 3,153 nt across 23 coding exons)